Lecture1

Monday, August 19, 2024 6:18 PM

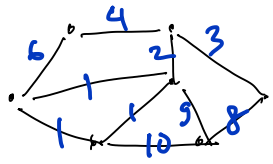
Minimum Spanning Trees (MST)

Graph: $G=(V,E)$

$w: E \to \mathbb{R}$

$|E| = m$

$|V| = n$



Spanning Tree

① Subset of edges that connect the vertices of the graph and is of size $V-1$

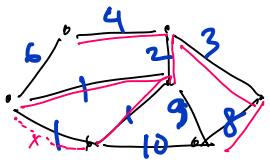
② Connected subgraph with no cycles.

MST: Minimum weighted subgraph satisfying Property ① or ②.

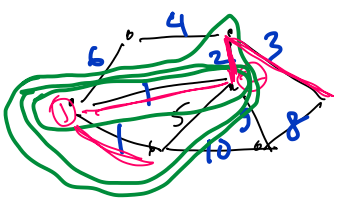
Algorithms for MST

Kruskal's Algorithm =

1. Sort edges in increasing order by weight.
   $e_1 < e_2 - - - - - < e_m$
2. $T \leftarrow \phi$
3. For $i = 1$ to m
   do
   if $T + \{e_i\}$ is a tree (does not form a cycle)
   $T \leftarrow T + \{e_i\}$



Prim's Algorithm:



$T \leftarrow \phi$; $S \leftarrow v_1$

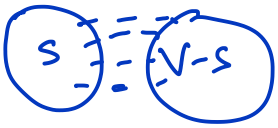
While $|S| < |V| = n$

do

$e \leftarrow$ least cost edge from S to $V-S$

$T \leftarrow T + \{e\}$

$S \leftarrow S + \{v\}$ $\quad e = (u,v)$ $u \in S, v \in V-S$.

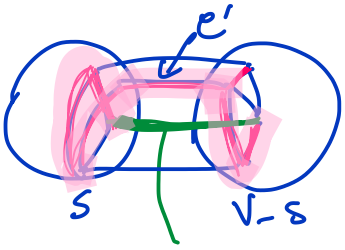


Cut-set$(S, V-S)$ = edges the removal of which partition the graph into at least 2 components.

Property

Let e_m be the minimum weighted edge in the cut-set Cut-Set$(S, V-S)$

Then $\exists$ a MST, M, s.t. $e_m \in$ MST M



$w(e_m) \leq w(e')$

Replace e' by e_m.

Proof (idea) of Prim's Algorithm.

In Prim the minimum weight edge in the cut-set $(S, V-S)$ is picked.

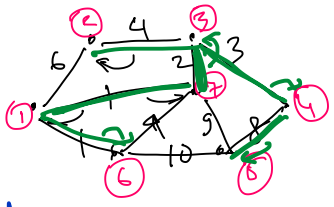
Proof idea of Kruskal's Algorithm

A new edge added by Kruskals

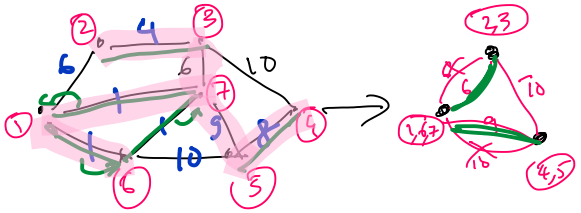


S V-S

Boruvka & Sollin :



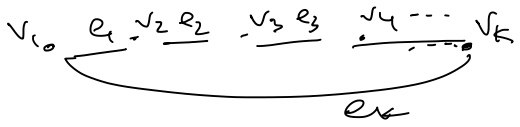
Repeat
1. Find minimum edge incident to every vertex



2. Find connected components.
   If ∃ a cycle remove any edge
3. Shrink connected components into one supervertex

Until # of components = 1 .

Property: All edge in the cycle have the same weight .



$w(e_1) \geq w(e_2) \geq w(e_3) \geq w(e_4) \geq \dots \geq w(e_k) \geq w(e_1)$